

Promised topic: The second summer school at BOUN will focus on advanced computational and multimodal approaches in magnetic resonance imaging (MRI). The program will provide participants with both theoretical foundations and hands-on insights into cutting-edge imaging techniques and data analysis methods used in contemporary neuroimaging and clinical research, including quantitative MRI techniques and MR spectroscopic imaging, along with deep learning.

Participants: MSc/PhD students, post-docs. 30 students accepted.

Date: June 29 - July 1, 2026, Boğaziçi University, Istanbul, Türkiye

Lecture room: VB 200 Google Maps <https://maps.app.goo.gl/zFyu8ryJ7WK3AKV36>
Video map for the VB200 from the entrance of the Institute for Data Science and Artificial Intelligence:
https://drive.google.com/file/d/1d6uN680Mwb3FdJgxPAsb4_-ZzTnUMmGu/view?usp=sharing

Confirmed speakers and assistants:

- Esin Öztürk Işık
- Pınar Özbay
- Arda Canbaş
- Melisa Özakçakaya
- Tolga Çukur (online)
- Abdullah Baş
- Said Aldemir
- Ayşe İrem Çetin
- Gülce Turhan
- Belal Tavashi
- Leen Hakki

Time	Topic		Speaker
First day: Magnetic resonance spectroscopic imaging, deep learning in the afternoon			
DAY 1			
9:00-9:15	Welcome and Introduction		Esin Öztürk Işık
9:15-10:30	Magnetic Resonance Spectroscopic Imaging (MRS/I)	Background and spectral structure components	Esin Öztürk Işık
10:30-10:45	Coffee break		
10:45-11:35	Proton and Carbon MRS	¹ H and ¹³ C combined for	Esin Öztürk Işık

		structure determination	
11:35-14:30	Graduation Ceremony+Lunch break		
14:30-15:30	LCModel and Oryx-MRSI	Spectroscopic data processing and analysis, practicals	Arda Canbaş and Melisa Özakçakaya
15:30-16:00	Clinical Applications of MRSI	How MRS/I is used in the clinics, metabolites	Esin Öztürk Işık
16:00-16:30	Introduction to Deep Learning	Deep learning algorithms, research at CIL using DL	Esin Öztürk Işık
16:30-17:15	Deep Learning - generative AI (GANs)	The curse of dimensionality	Klaus Eickel
Second day morning: Deep learning and Quantitative MRI			
DAY 2			
Day 2 9:00-10:15	Deep learning for image classification, segmentation, super resolution and VLMS		Said Aldemir / Abdullah Baş
10:15-10:45	Coffee break		
10:45-11:30	Deep learning for MRI reconstruction		Tolga Çukur (online)
11:30-12:30	Deep learning practicals		Abdullah Baş, Said Aldemir, Ayşe İrem Çetin, Gülce Turhan
12:30-13:30	Lunch break		
Second day afternoon: Quantitative MRI			
13:30-14:30	From Contrast-Weighted Imaging to Quantitative		Pınar S Özbay

	Mapping: Multiparametric MRI		
14:30-15:00	Coffee break		
15:00-16:30	Relaxometry in Human and Preclinical MRI - Creating T1, T2, T2*, and R2* Maps	Hands-on Tutorial I	Leen Hakki, Belal Tavashi
17:30-20:00	Dinner (for speakers)		
DAY 3			
Third day morning: Quantitative MRI (continue)			
9:00-10:30	Susceptibility-Based MRI: From SWI to QSM, with Clinical and Preclinical Applications		Pınar S Özbay
10:30-11:00	Coffee break		
11:00-12:30	SWI/QSM Processing, plus a Deep Learning DEMO	Hands-on Tutorial II	Belal Tavashi and Leen Hakki
12:30-13:30	Lunch		

- Time slots are subject to change.

Instructions for the participants

MRSI Part:

- 1- Preparation: Please take a look at the basics of organic chemistry.
- 2- Download [Docker Desktop](#)
- 3 - Please use the following instructions to set up your computer before the Summer School.

Windows [instructions](#)

MACOS [instructions](#)

Linux [instructions](#)

- 4) Structure Determination using 1H MRS and 13C MRS [Practical](#)

Deep Learning Part:

No need for specific installations. We will ask you to use Google Colab, so you might need a Google account.

qMRI Part:

Please follow the [instructions](#) for the qMRI part.

Participants are expected to have access to:

- A personal laptop
- A working **Python environment**
- **MATLAB**
- **SEPIA toolbox** installed and tested before the hands-on session
- Basic familiarity with MRI image formats and data handling is helpful but not required